PROIECT DEEA 2022-2023

NUME: Gîrniță Alexandra-Claudia

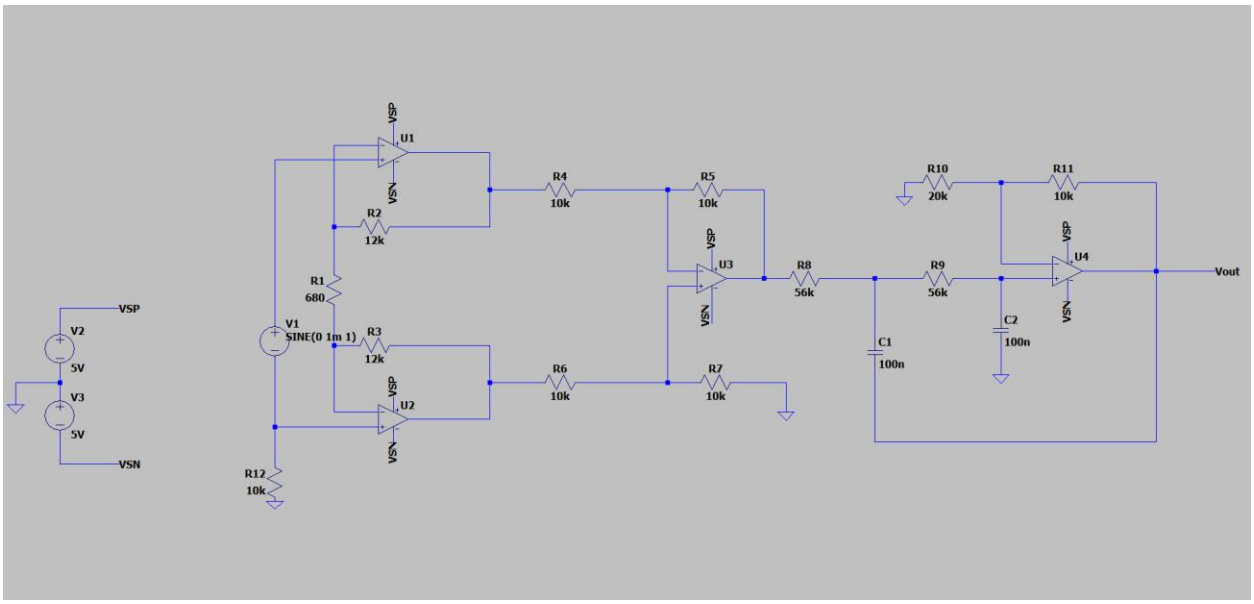
Grupa: 322CC

Cerinte-analiza:

1. Calcularea valorilor numerice:

N	U	M	E		P	R	E	N	U	M	E
G	Î	R	N	IȚĂ	A	L	E	X	A	N	DRA
L1	L2	L3			L4	L5	L6				
680	25	56			12	3	500				
R1 (Ω)	V_{im} (mV) intrare	R8=R9 (kΩ)			R2=R3 (kΩ)	V_{om} (V) iesire	f_{-3dB} (Hz) banda				

- Schema in LTSpice cu valorile numerice personalizate:



- Am folosit pentru proiectare amplificatorul operational: universalOpamp2.

2. SIMULAREA DE TIP DC SWEEP

Calcule teoretice(analiza pe blocuri)

- Formula de la amplificatorul neinversor: $1 + \frac{R_2+R_3}{R_1} = 1 + \frac{12k+12k}{680} = \frac{617}{17} = 36.3$
- Diferential: $-\frac{R_5}{R_4} = -\frac{10k}{10k} = -1$
- Neinersor: $1 + \frac{R_{11}}{R_{10}} = 1 + \frac{10k}{20k} = 1.5$

Amplificarea de tensiune: $A = \left(1 + \frac{R_2+R_3}{R_1}\right) * \left(-\frac{R_5}{R_4}\right) * (1 + \frac{R_{11}}{R_{10}})$

$$A = 36.3 * (-1) * 1.5 = -54.45$$

Domeniul tensiunii de intrare (schema functioneaza linear): ➔ [-91.9; 91,9] mV

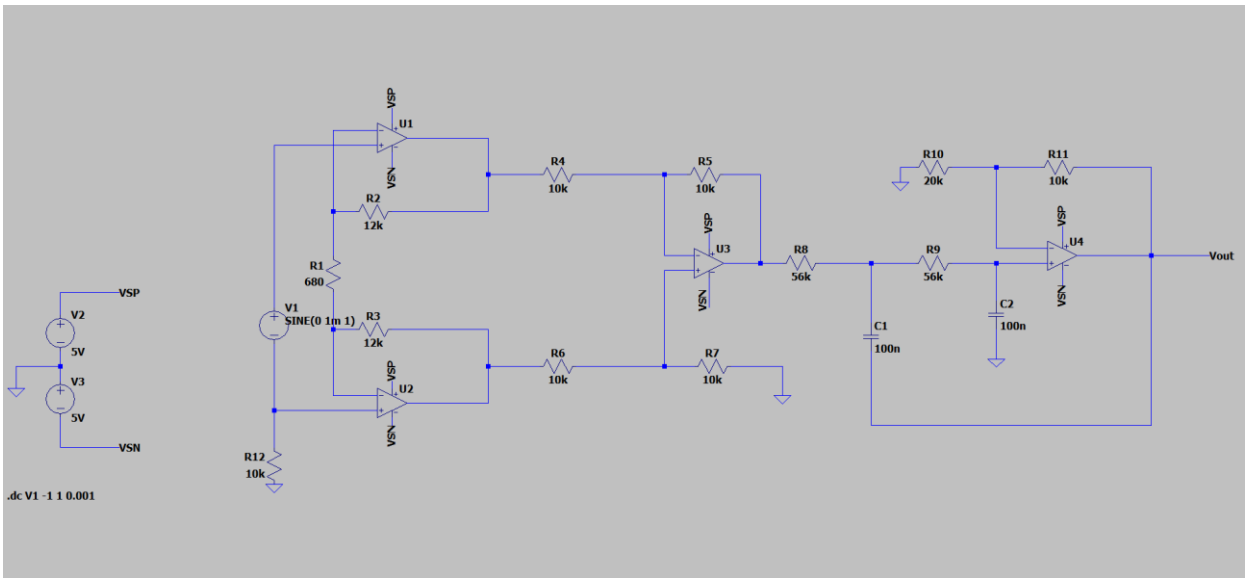
- $V_{in_minim} = \frac{-5}{|-54.45|} = -0.0919V$
- $V_{in_maxim} = \frac{5}{|-54.45|} = 0.0919V$

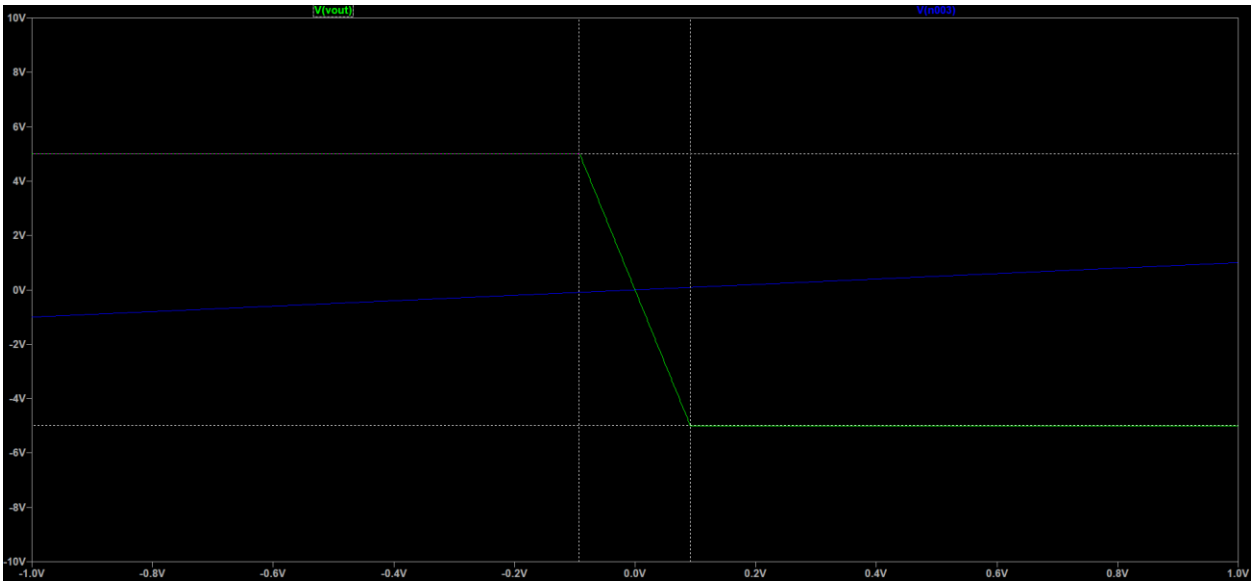
Domeniul tensiunii de iesire (valori intre VSP si VSN)

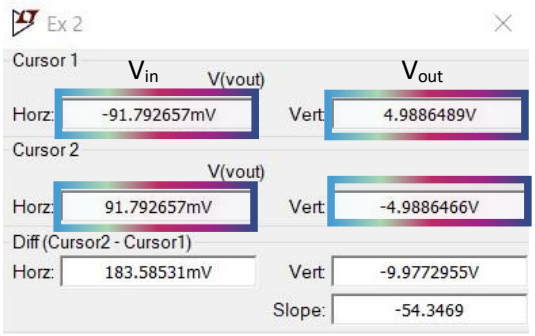
- $V_{out_minim} = -5V$
- $V_{out_maxim} = 5V$

Pentru simularea in LTSpice, am ales variatia tensiunii V_1 intre -1V si 1V cu pas de 1mV.

Comanda simulare: .dc V1 -1 1 0.001







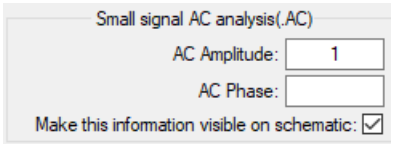
- Amplificarea conform graficului : -54.3469
- Amplificarea teoretica:

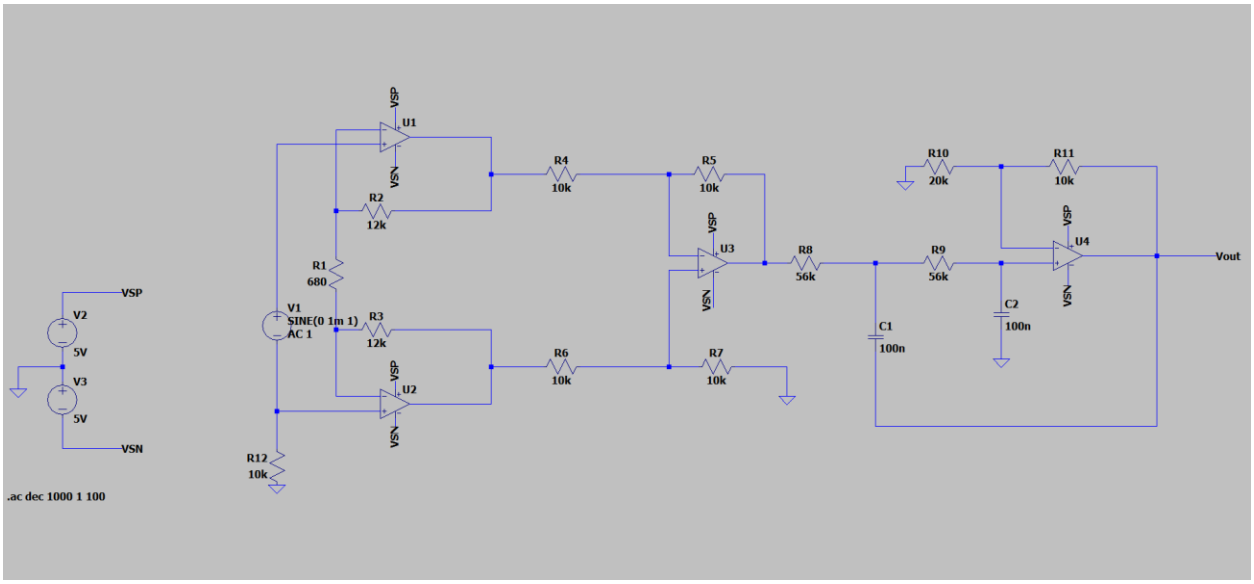
$$A = \frac{V_{out}}{V_{in}} = \frac{-4.99 * 10\text{^}3}{91.79} = -54.45$$

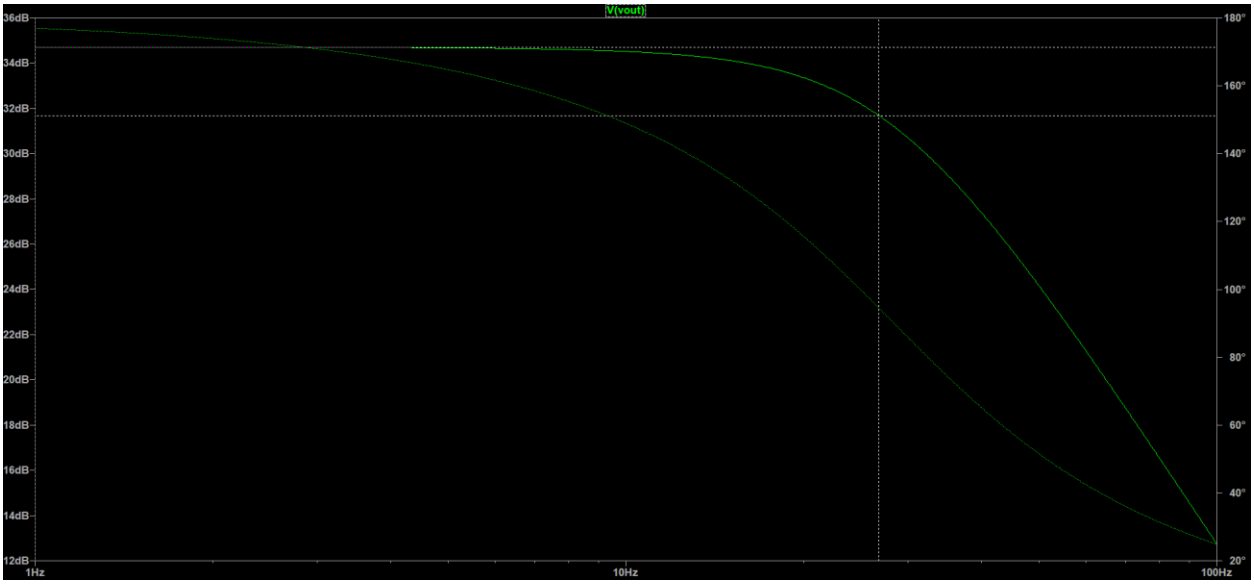
- Am plasat cursoarele la distanta egala in modul unul fata de celalalt astfel incat sa obtin domeniul liniar cuprins intre cele doua [-91.9; 91,9] mV. Astfel am obtinut o amplificare de aproximativ -54.3, aceasta fiind negativa datorita sensului in care am parcurs schema.
- Dupa efectuarea calculelor de mana a rezultat un raspuns asemanator.
- Abatarea de 0.018% se datoreaza faptului ca amplificatorul simuleaza o piesa reala si are amplificarea in bucla deschisa finita

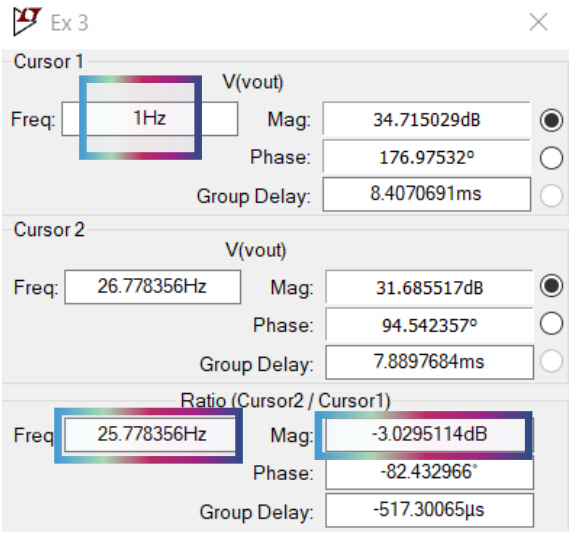
3. SIMULARE DE TIP AC

- Comanda simulare: .ac dec 1000 1 100
- Modificam valoarea sursei :





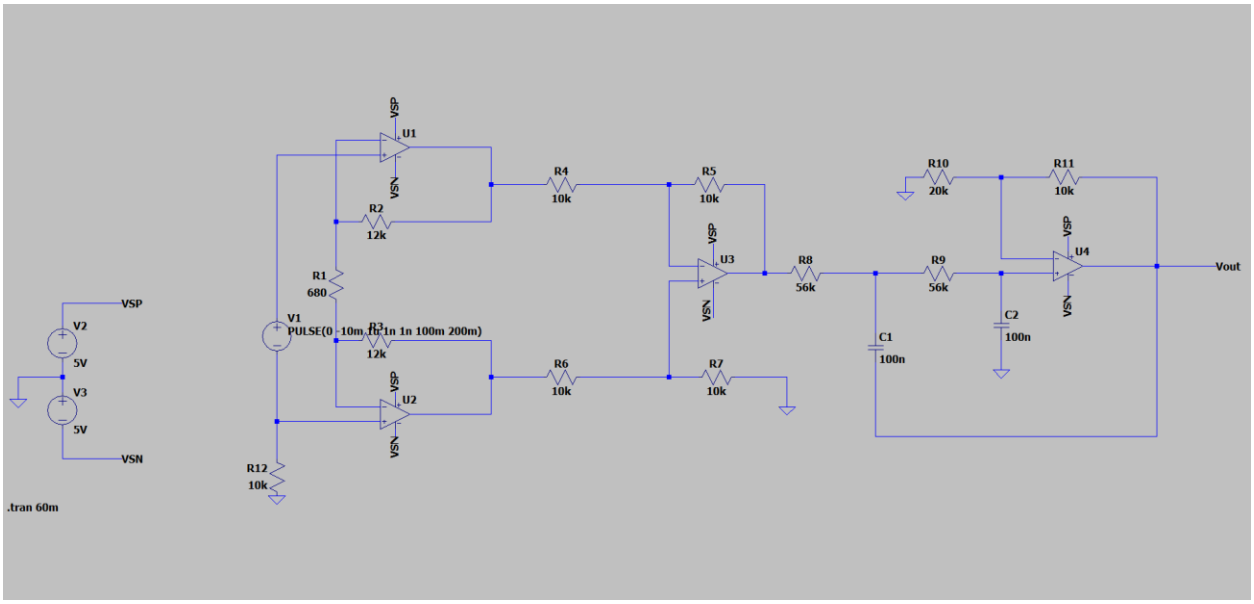


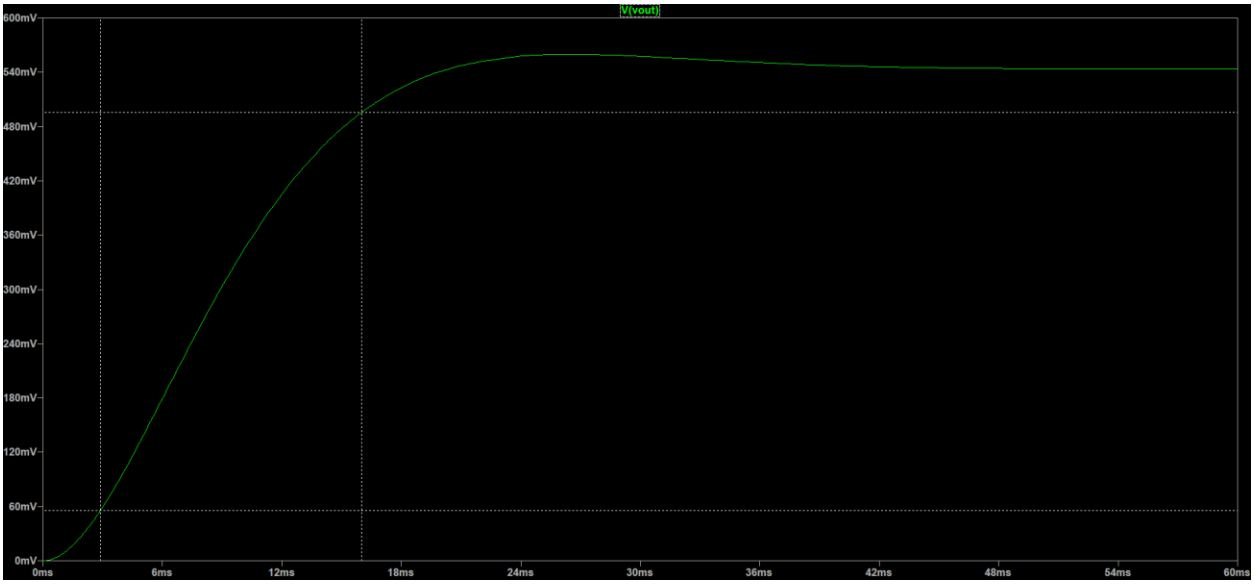


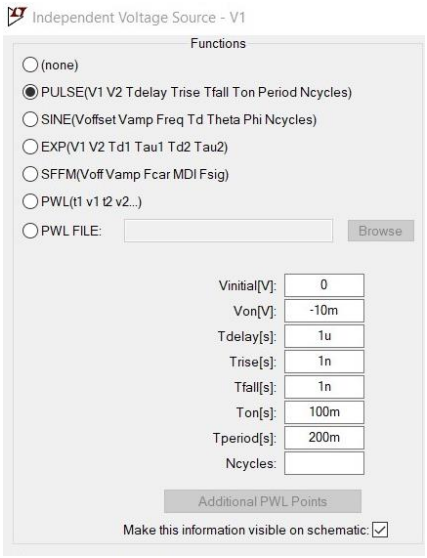
- Primul cursor la 1Hz -> o frecventa cat mai exacta
- Al doilea cursor l-am amplasat astfel inca sa avem acea scadere a amplificarii cu 3dB
- Frecventa rezultata din simulare 25.77 Hz la -3dB
- Caracteristica de frecventa a schemei reiese din grafic
- Linia subtire -> defazajul
- Linia proeminenta -> amplificarea

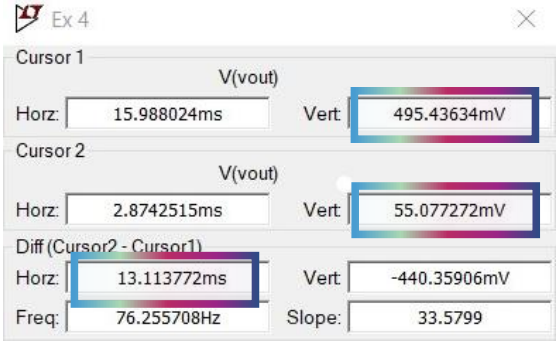
4. SIMULARE DE TIP TRANSIENT

- Comanda simulare: .trans 60m
- Timpul de crestere ➔ 13.113 ms









- Frecventa calculate anterior(exercitiul 3) => 26.77Hz

- Perioada => $\frac{1}{26.77Hz} = 0.037355\ s = 37.35\ ms$

$$\frac{37.35\ ms}{13.11\ ms} = 2.84 \quad ==>$$

perioada aferentă frecvenței caracteristice filtrului este de 2.84 de ori mai mare decât timpul de creștere.

Cerinte-proiectare:

5. Modificare schemei initiale

5.1 Modificarea schemei initiale pentru o anumita amplificare

$V_{im} = 25mV$ => domeniul de intrare : [-25; 25] mV

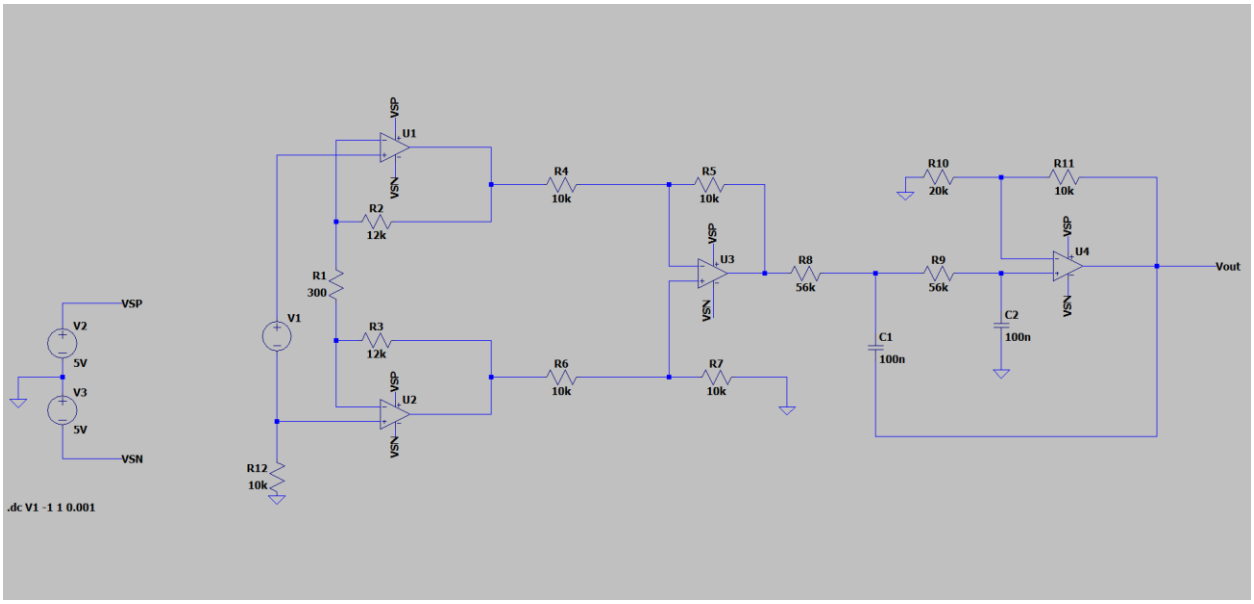
$V_{out} = 3V$=> domeniul de intrare : [-3; 3] V

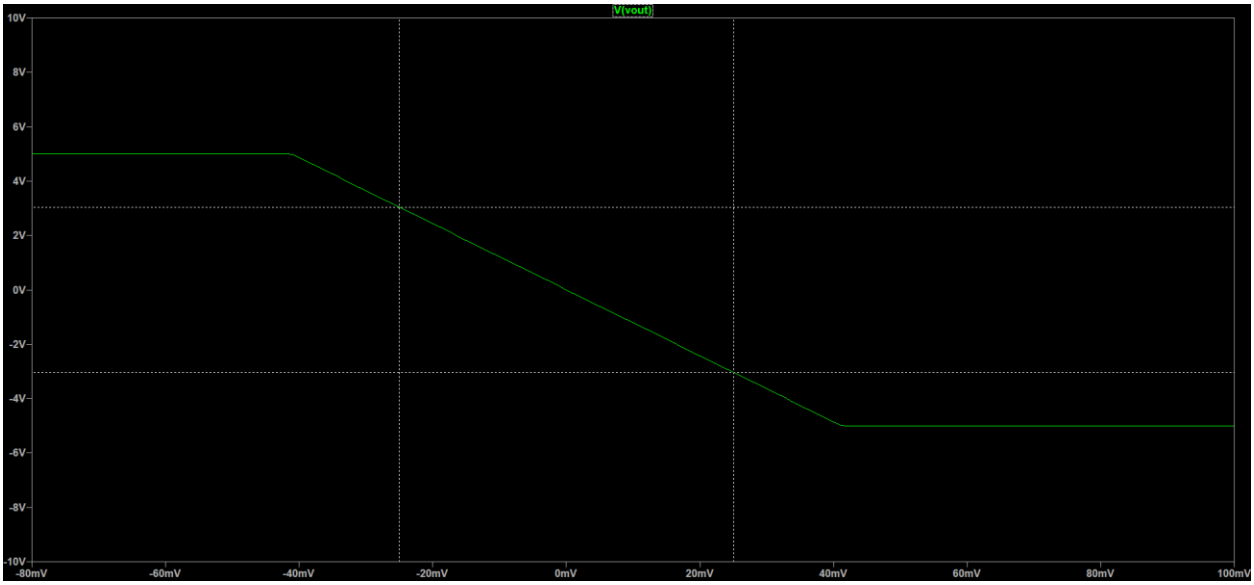
$$A = \frac{V_{out}}{V_{im}} = \frac{3}{-0.025} = -120 ==> A = -120$$

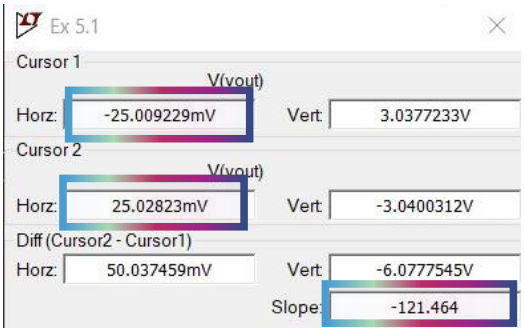
$$A = \left(1 + \frac{R_2 + R_3}{R_1}\right) * \left(-\frac{R_5}{R_4}\right) * (1 + \frac{R_{11}}{R_{10}})$$

$$-120 = \left(1 + \frac{12k + 12k}{R_1'}\right) * \left(-\frac{10k}{10k}\right) * \left(1 + \frac{10k}{20k}\right) ==> R_1' = 303\Omega$$

Standard E24 : $R_1 = 300\ \Omega$







Eroarea:

$$A = -121.464$$

$$\left(1 - \frac{-120}{-121.464}\right) * 100 = 1.2\% < 10\%$$

5.2 Modificarea schemei initiale pentru frecventa de -3dB

$$f_{-3dB} = 500\ Hz$$

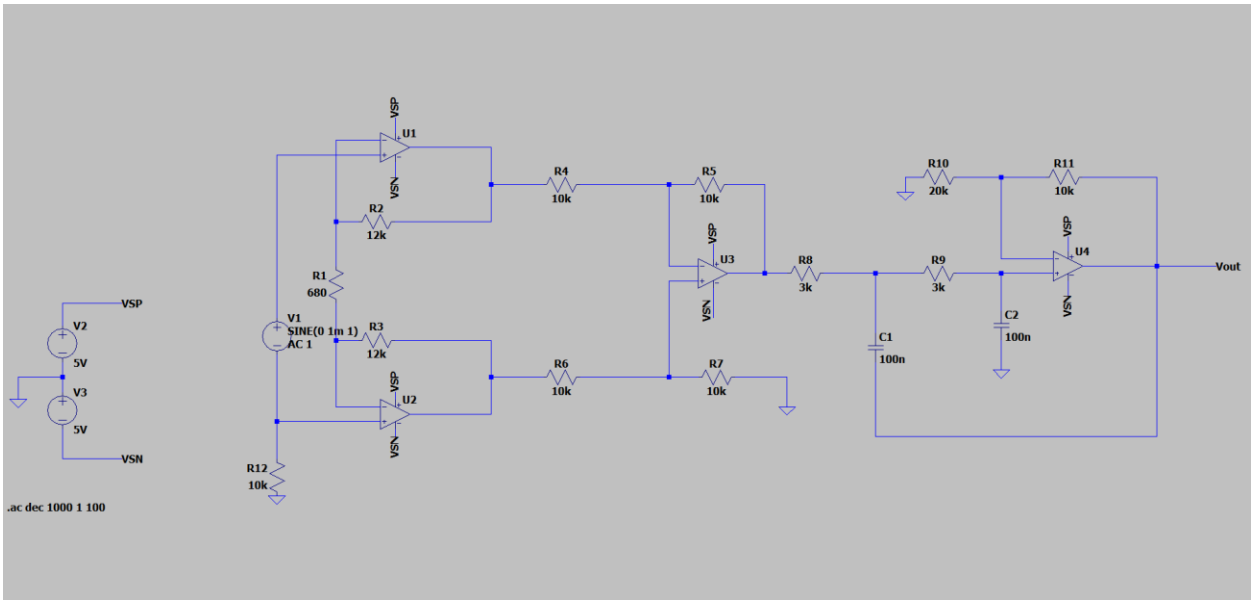
$$A_{fv} = \frac{500}{26.77} = 18.67 ==> amplificarea\ frecventei\ de - 3dB\ de\ 18.67\ ori$$

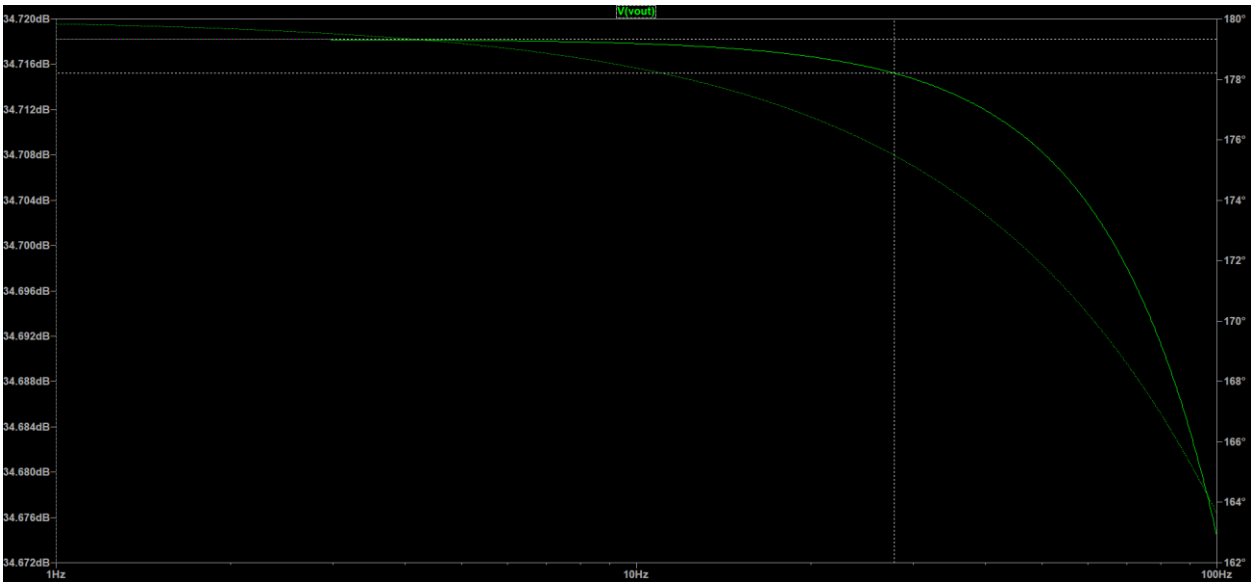
$$\frac{1}{\sqrt{R_8 R_9 C_1 C_2}} = 2\pi f_{-3dB}$$

$$==>\ trebuie\ scazute\ fie\ cele\ 2\ rezistente\ ,\ fie\ cele\ 2\ condensatoare$$

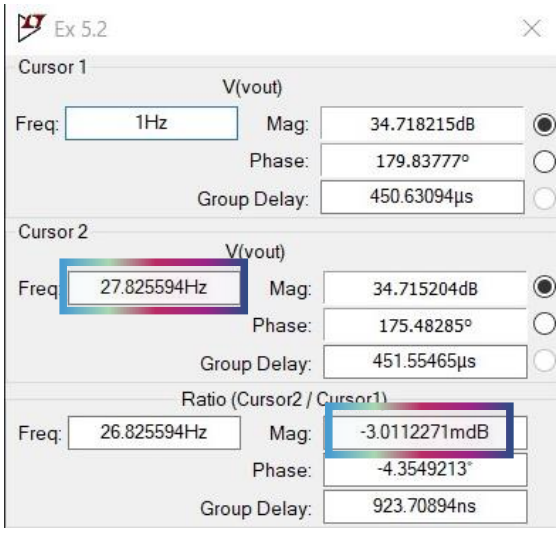
$$R_8 = R_9 = \frac{56k}{18.67} = 2999.46\ \Omega \cong 3\text{k}\Omega$$

Standard E24 : $R_8 = R_9 = 3\text{k}\Omega$





- Frecventa => 27.82 Hz
- Eroarea => $\left(1 - \frac{26.77Hz}{27.82Hz}\right) * 100 = 3.77\% \ < 10\%$



Concluzii finale:

- Schema este un amplificator de instrumentatie. Citeste o tensiune electrica venita de la un sensor care e controlat cu sursa V1. Tensiunea respectiva este amplificata. Semnalul(V1) nu este masurat fata de masa, ci intre cele doua intrari, intre care am legat si senzorul.
- Filtrul activ trece-jos limiteaza componentele rapide din semnalul de iesire (atenueaza frecventele inalte).
- Analiza de tip DC Sweep ne permite o analiza de precizie mai buna decat analiza de tip Transient. In DC sweep nu mai exista raspuns in frecventa (inexistenta condensator).
- Analiza de tip AC ne da caracteristica de frecventa datorita existentei unui filtru.
- Aceasta schema preia un semnal mic diferential si il transforma intr-un semnal mare fata de masa, filtrat.